Bangladesh University of Business & Technology

Rupnagar, Mirpur-2, Dhaka-1216

BUBT CANTEEN FOOD ORDERING SYSTEM (BUBT CFOS)

UNDERGRADUATE PROJECT

Submitted in partial fulfillment of the requirements of Software Development project 1 for the degree of B.Sc. Engg. In CSE by Intake-50, Section-5, Group-1.

Prepared By:

Samia Rahman
22234103338
Asha Haider Heeya
22234103340
Masum Billah
22234103343
Lamia Binte Momin
22234103347
Shahadat Hossain
22234103356

Supervisor:

Md. Mahbub-Or-Rashid
Assistant Professor

Department of CSE, BUBT

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Declaration of Authorship

We declare that this project titled "**BUBT CANTEEN FOOD ORDERING SYSTEM (BUBT CFOS)**"and the work presented in it are our own.

We, hereby declare that this submission is entirely our own work, in our own words, and that all sources used in researching it are fully acknowledged and all quotations properly identified. We are aware that this offline project of ours published in digital form can be used by everyone without using internet. It has not been submitted in whole by us for the purpose of obtaining any other credit or grade. We understand the ethical implications of our research and this work meets the requirements of the Faculty of Computer Science Engineering.

Name: Samia Rahman
Id: 22234103338
Name: Asha Haider Heeya
Id: 22234103340
Name: Masum Billah
Id:22234103343
Name: Lamia Binte Momin
Id:22234103347
Name: Shahadat Hossain
Id: 22234103356

Certificate

This is to certify that the project entitled, "**BUBT CANTEEN FOOD ORDERING SYSTEM (BUBT CFOS)**" and submitted by **Lamia Binte Momin** (22234103347), **Asha Haider Heeya** (22234103340), **Samia Rahman** (22234103338), **Shahadat Hossain** (22234103356), **Masum Billah** (22234103343) in partial fulfillment of the requirements of embodies the work done by them under my supervision.

Md. Mahbub-Or-Rashid
Assistant Professor, CSE, BUBT

Dedication

Dedicated to our parents for all their love and inspiration.

Abstract

The BUBT Canteen Food Ordering System is a software application designed to streamline the ordering process at the canteen of **Bangladesh University of Business and Technology**. The system is implemented in C programming language, and it provides an easy-to-use interface for customers to place orders, and for staff to manage and fulfill those orders.

This report outlines the development of the BUBT Canteen Food Ordering System, including its objectives, design, and implementation. The system was developed in response to the inefficiencies and manual processes at the BUBT canteen, which led to long wait times and errors in order processing.

The report describes the methodology used in the development of the system, including the tools and techniques used, and the challenges faced during the development process. It also discusses the features of the system, including its user interface Overall, the BUBT Canteen Food Ordering System has the potential to significantly improve the ordering process at the BUBT canteen, and to provide a more efficient and streamlined experience for students and staff alike.

Acknowledgments

First of all, we are thankful and expressing our gratefulness to Almighty Allah who offers us His divine blessing, patient, mental and physical strength to complete this project work. We are deeply indebted to our project supervisor Md. Mahbub-Or-Rashid, Assistant Professor, Department of Computer Science and Engineering (CSE), Bangladesh University of Business and Technology (BUBT). His scholarly guidance, important suggestions, work for going through our drafts and correcting them, and generating courage from the beginning to the end of the research work has made the completion of this thesis possible. A very special gratitude goes out to all our friends for their support and help to implement our works. The discussions with them on various topics of our works have been very helpful for us to enrich our knowledge and conception regarding the work. Last but not the least; we are highly grateful to our parents and family members for supporting us spiritually throughout writing this thesis and our life in general.

Approval

This project "**BUBT Canteen Food Ordering System**" Submitted by **Lamia Binte Momin** (22234103347), **Asha Haider Heeya** (22234103340), **Samia Rahman** (22234103338), **Shahadat Hossain** (22234103356), **Masum Billah** (22234103343), department of Computer Science and Engineering (CSE), Bangladesh University of Business and Technology (BUBT) under the supervision of Md. Mahbub-Or-Rashid Assistant professor, Department of Computer Science and Engineering has been accepted as satisfactory for the partial fulfillment of the requirement for the degree of Bachelor of Science (B.Sc. Engg.) in Computer Science and Engineering and approved as to its style and contents.

Supervisor:

Md. Mahbub-Or-Rashid
Assistant Professor
Department of CSE
Bangladesh University of Business and Technology (BUBT)

Chairman:

Dr. Muhammad Firoz Mrida
Associate Professor & Chairman
Bangladesh University of Business and Technology (BUBT)

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Chapter 1

Introduction

1.1 Introduction

This proposal is about building a software called "BUBT CFOS" for all students, faculty, and employees of BUBT. "BUBT Canteen Food Ordering System" is the full form of "BUBT CFOS." By using this software, every student at BUBT can order any food from each class room in the two canteens at BUBT and receive their ordered food in a very short time.
This System will be an interactive and up-to-date menu with all available options for all food items. Customer can choose one or more food items to place an order which will land in the Cart. And that Cart will contain all the ordered items for delivering the order to the customer. And Customers can view all the order details in the cart before checking out. In the end, the customer gets order confirmation details. Once the order will be placed it will enter the database system and retrieved that information. This System will allow Canteen Employees to rapidly go through the orders as they are received and process all orders efficiently for delivering and effectively with minimal delays and confusion to make happy the customers.

Chapter 2

Technologies

2.1 Software

2.1.1 Code :: Blocks

Code::Blocks is a free, open-source cross-platform IDE that supports multiple compilers including GCC, Clang and Visual C++. It is developed in C++ using wxWidgets as the GUI toolkit. Using a plugin architecture, its capabilities and features are defined by the provided plugins. Currently, Code::Blocks is oriented towards C, C++, and Fortran. It has a custom build system and optional Make support. Code::Blocks is being developed for Windows and Linux (the latest macOS version is 13.12 released on 12/26/2013) and has been ported to FreeBSD, OpenBSD and Solaris.

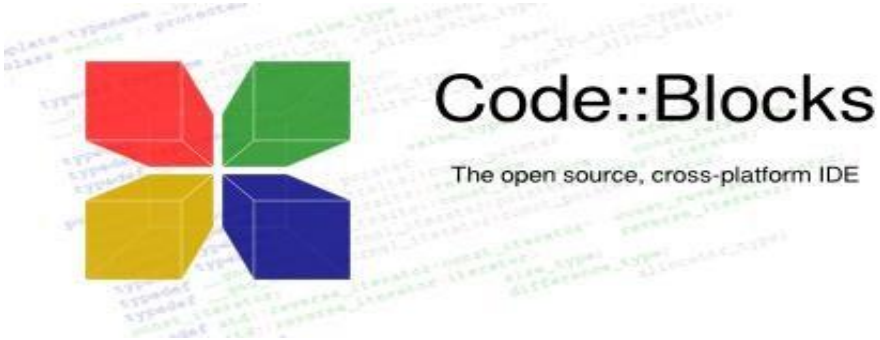


Figure 2.1: CodeB::locks

After releasing two release candidate versions, 1.0rc1 on July 25, 2005 and 1.0rc2 on October 25, 2005, instead of making a final release, the project developers started adding many new features, with the final release being repeatedly postponed. Instead, there were nightly builds of the latest SVN version made available on a daily basis. [citation needed] The first stable release was on February 28, 2008, with the version number changed to 8.02. The versioning scheme was changed to that of Ubuntu, with the major and minor number representing the year and month of the release. Version 17.12 is the latest stable release; however for the most up-to-date version the user can download the relatively stable nightly build or download the source code from SVN. Jennic Limited distributes a version of Code::Blocks customized to work with its microcontrollers. [Wikepedia]

2.2 Programming Language

2.2.1 C Language

C (/si/, as in the letter c) is a general-purpose, imperative computer programming language, supporting structured programming, lexical variable scope and recursion, while a static type

system prevents many unintended operations. By design, C provides constructs that map efficiently to typical machine instructions, and therefore it has found lasting use in applica- tions

that had formerly been coded in assembly language, including operating systems, as well as various application software for computers ranging from supercomputers to embedded systems.

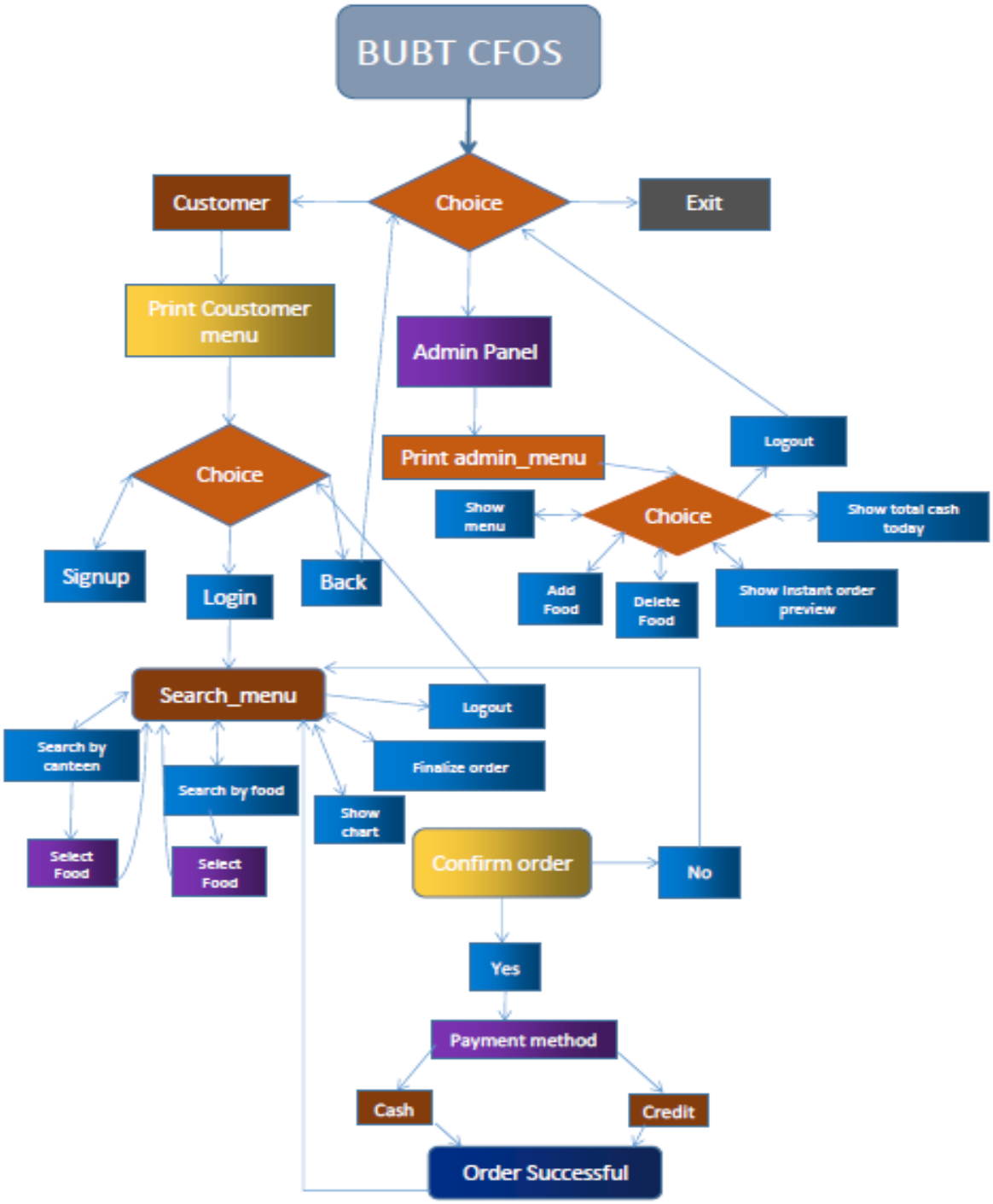
Figure 2.2: C Language

Many later languages have borrowed directly or indirectly from C, including C, C#, Unix's C shell, D, Go, Java, JavaScript, Limbo, LPC, Objective-C, Perl, PHP, Python, Rust, Swift, Verilog and System Verilog (hardware description languages). These languages have drawn many of their control structures and other basic features from C. Most of them (with Python being the most dramatic exception) are also very syntactically similar to C in general, and they tend to combine the recognizable expression and statement syntax of C with underlying type systems, data models, and semantics that can be radically different. [Wikipedia]

Chapter 3

System Analysis & Architectural Design

3.1 Flowchart:



Chapter 4

Access System

4.1 Description:

By reading this chapter user can know about every options of this system as like, how to operate this system, who can operate this system and all restrictions.

4.2 Manual:

At first run, we can see a page like Figure-4.1 and it asks to select an option.

```
        ********Welcome to BUBT Canteen Food Ordering System(CFOS)*******

Please select an option:

        1) Customer
        2) Admin Panel
        3) Exit

Enter your choice:
```

Figure - 4.1

1. If we select option 1 then it will show a page like Firgure - 4.2.

```
        ****Customer Menu****

Please select an option:

        1) Signup
        2) Login
        3) Back

Enter your choice:
```

Figure - 4.2

2. If I don't have an account then I have to select option 1 as shown in Fig-4.2. To create a account I need to enter my name, email, mobile number, and password as shown in Fig - 4.3. If my name,

email, mobile number, and password doesn't satisfy the conditions, it will show an error. Signup will be successful if input satisfy the conditions.

```
        ****Signup****

Enter your name: ash2
        Invalid name! Please enter a name with alphabets only.
        Enter your name: asha
Enter your email: asha23
Invalid email!(Email must contain @, a dot and length should be greater than 5.)
Enter your email: asha@gmail.com
Enter your mobile number: 123456789
Invalid mobile number! Please enter a valid 11-digit number.
Enter your mobile number: 12345678923
Enter a password: 2345
Password length must be between 5 to 12 characters.
Invalid password! Please enter a valid password.
Enter a password: 23456


Signup successful! Please login to continue.

        Press any key to continue...
```

Figure - 4.3

3. If I had an account before then I can login with my registered email and password like figure - 4.5. If my email or password is wrong then it will show an error like figure - 4.4.

```
        ****Login****

Enter your email: heeya
Enter your password: 1234

Incorrect email or password
```

Figure - 4.4

```
        ****Login****

Enter your email: heeya@
Enter your password: 1234

Login successful. Welcome heeya!
```

Figure - 4.5

4. After login, there is a page like Figure - 4.6 where I can select an option. If I select option 1 then there will be a page like Figure - 4.7 where I can select a canteen.

```
****Search Menu****

Please select an option:

1) Search by Canteen
2) Search by Food
3) Show Cart
4) Finalize Order
5) Logout

Enter your choice:
```

Figure - 4.6

```
        Search by Canteen

Please select an option:
        1) Building-3 Canteen
        2) BBA Canteen
        3) Back
Enter canteen number:
```

Figure - 4.7

5. If I choose option 1, there will be a page like Figure - 4.8 where I will see the food item, quantity and price.

```
                Welcome to BUBT Building-3 Canteen
  Food Code        Name                          Quantity       Price
  ---------------------------------------------------------------------
     1         |  Chicken Burger                 |    20      |  50.00
  ---------------------------------------------------------------------
     2         |  Burger                         |    16      |  55.00
  ---------------------------------------------------------------------
     3         |  Chicken Roll                   |    22      |  35.00
  ---------------------------------------------------------------------
     4         |  Sandwich                       |    25      |  45.00
  ---------------------------------------------------------------------
     5         |  Pizza                          |    19      |  70.00
  ---------------------------------------------------------------------
     6         |  Chicken nugget                 |    34      |  40.00
  ---------------------------------------------------------------------
     7         |  Ice cream                      |    24      |  50.00
  ---------------------------------------------------------------------
     8         |  Noodles                        |    16      |  55.00
  ---------------------------------------------------------------------
     9         |  Chips                          |    20      |  20.00
  ---------------------------------------------------------------------
    10         |  French Fries                   |    15      |  100.00
  ---------------------------------------------------------------------
    11         |                                 |     0      |  0.00
  ---------------------------------------------------------------------
    12         |                                 |     0      |  0.00
  ---------------------------------------------------------------------
    13         |                                 |     0      |  0.00
  ---------------------------------------------------------------------
    14         |                                 |     0      |  0.00
  ---------------------------------------------------------------------
    15         |                                 |     0      |  0.00
  ---------------------------------------------------------------------
    16         |                                 |     0      |  0.00
  ---------------------------------------------------------------------
    17         |                                 |     0      |  0.00
  ---------------------------------------------------------------------
    18         |                                 |     0      |  0.00
  ---------------------------------------------------------------------
    19         |                                 |     0      |  0.00
  ---------------------------------------------------------------------
    20         |                                 |     0      |  0.00
  ---------------------------------------------------------------------
Enter item number to add to cart (enter 0 to cancel):
```

Figure - 4.8

6. I need to enter the item number and the quantity to add into the cart. But if the item number is wrong and the quantity doesn't match the card then an error will occur.

```
Enter item number to add to cart (enter 0 to cancel): 3
Enter quantity: 9
Item added to cart

Enter item number to add to cart (enter 0 to cancel): 4
Enter quantity: 5
Item added to cart

Enter item number to add to cart (enter 0 to cancel): 7
Enter quantity: 25
Sorry, there is not enough quantity of the selected food item.

Enter item number to add to cart (enter 0 to cancel): 7
Enter quantity: 10
Item added to cart

Enter item number to add to cart (enter 0 to cancel): 19
Enter quantity: 1
Sorry, there is not enough quantity of the selected food item.

Enter item number to add to cart (enter 0 to cancel):
```

Figure - 4.9

7. If I select option 2 as shown in Figure - 4.7, there will be a page like Figure - 4.10 where I can see different types of food items, quantities, and prices.

```
                Welcome to BUBT BBA Canteen
  Food Code        Name                          Quantity       Price
  ---------------------------------------------------------------------
    1       |  Burger                            |    20       |  50.00
  ---------------------------------------------------------------------
    2       |  Khichuri                          |    16       |  55.00
  ---------------------------------------------------------------------
    3       |  Chicken Roll                      |    22       |  45.00
  ---------------------------------------------------------------------
    4       |  Sandwich                          |    25       |  45.00
  ---------------------------------------------------------------------
    5       |  Pizza(Family size)                |    18       |  70.00
  ---------------------------------------------------------------------
    6       |  Ice cream                         |    26       |  45.00
  ---------------------------------------------------------------------
    7       |  Beef Burger                       |    19       |  50.00
  ---------------------------------------------------------------------
    8       |  French Fries                      |    19       |  100.00
  ---------------------------------------------------------------------
    9       |  Club Sandwich                     |    20       |  80.00
  ---------------------------------------------------------------------
   10       |  Cake                              |    35       |  10.00
  ---------------------------------------------------------------------
   11       |                                    |     0       |  0.00
  ---------------------------------------------------------------------
   12       |                                    |     0       |  0.00
  ---------------------------------------------------------------------
   13       |                                    |     0       |  0.00
  ---------------------------------------------------------------------
   14       |                                    |     0       |  0.00
  ---------------------------------------------------------------------
   15       |                                    |     0       |  0.00
  ---------------------------------------------------------------------
   16       |                                    |     0       |  0.00
  ---------------------------------------------------------------------
   17       |                                    |     0       |  0.00
  ---------------------------------------------------------------------
   18       |                                    |     0       |  0.00
  ---------------------------------------------------------------------
   19       |                                    |     0       |  0.00
  ---------------------------------------------------------------------
   20       |                                    |     0       |  0.00
  ---------------------------------------------------------------------

Enter item number to add to cart (enter 0 to cancel):
```

Figure - 4.10

8. Here also I need to enter the item number and quantity to add into the cart. This will do the same as in Figure 4.9 earlier.

```
Enter item number to add to cart (enter 0 to cancel): 2
Enter quantity: 4
Item added to cart

Enter item number to add to cart (enter 0 to cancel): 10
Enter quantity: 10
Item added to cart

Enter item number to add to cart (enter 0 to cancel): 9
Enter quantity: 9
Item added to cart

Enter item number to add to cart (enter 0 to cancel): 11
Enter quantity: 3
Sorry, there is not enough quantity of the selected food item.

Enter item number to add to cart (enter 0 to cancel):
```

Figure - 4.11

9. After selecting the food, I can see the total price of the food in the cart, the discount, and the total bill after the discount if I select option 3 as shown in Figure 4.6.

```
Cart Contents:


  Name                        Quantity      Price      Total
-----------------------------------------------------------------------
  Chicken Roll                |     9      |  35.00   |  315.00
-----------------------------------------------------------------------
  Sandwich                    |     5      |  45.00   |  225.00
-----------------------------------------------------------------------
  Ice cream                   |    10      |  50.00   |  500.00
-----------------------------------------------------------------------
  Khichuri                    |     4      |  55.00   |  220.00
-----------------------------------------------------------------------
  Cake                        |    10      |  10.00   |  100.00
-----------------------------------------------------------------------
  Club Sandwich               |     9      |  80.00   |  720.00
-----------------------------------------------------------------------
  Total                                                  : 2080.00
-----------------------------------------------------------------------
  Discount: 208.00
-----------------------------------------------------------------------
  Total bill (after 10% discount): 1872.00
-----------------------------------------------------------------------
```

Figure - 4.12

10. After seeing the price, now I need to confirm my order. If I select yes then I have to enter my building and room number. After that, I have to select a payment method between cash and credit. If I select cash my order will be confirmed successfully. Then I have to select option 5 to exit.

```
Do you want to confirm your order?
        1) Yes
        2) No
Enter your choice: 1
        Enter your building number(B1/ B2/ B3/ B4) : B2
        Enter your room number : 505

Payment Method:
        1.Cash
        2.Credit
        1

        =========>THANK YOU<=========


The order has been placed successfully.

****Search Menu****

Please select an option:

1) Search by Canteen
2) Search by Food
3) Show Cart
4) Finalize Order
5) Logout

Enter your choice:
```

Figure - 4.13

11. If I log in to another account then I will see the quantity of food changes.

```
        Search by Canteen

Please select an option:
        1) Building-3 Canteen
        2) BBA Canteen
        3) Back
Enter canteen number: 2

                Welcome to BUBT BBA Canteen

          Food Code       Name                        Quantity      Price
        ---------------------------------------------------------------------
           1      |  Burger                          |    20      |  50.00
        ---------------------------------------------------------------------
           2      |  Khichuri                        |    12      |  55.00
        ---------------------------------------------------------------------
           3      |  Chicken Roll                    |    22      |  45.00
        ---------------------------------------------------------------------
           4      |  Sandwich                        |    25      |  45.00
        ---------------------------------------------------------------------
           5      |  Pizza(Family size)              |    18      |  70.00
        ---------------------------------------------------------------------
           6      |  Ice cream                       |    16      |  45.00
        ---------------------------------------------------------------------
           7      |  Beef Burger                     |    19      |  50.00
        ---------------------------------------------------------------------
           8      |  French Fries                    |    19      |  100.00
        ---------------------------------------------------------------------
           9      |  Club Sandwich                   |    11      |  80.00
        ---------------------------------------------------------------------
          10      |  Cake                            |    25      |  10.00
        ---------------------------------------------------------------------
```

Figure - 4.14

12. If I select option 2 in Figure 4.6 then there will be a page like Figure - 4.15 where I can enter a food name. If I can't write the correct name then there will be shown an error as like Figure - 4.15. After writing the correct name of the food there will show the food, quantity, price, and canteen name as like Figure - 4.16.

```
        Search by Food

Enter food name: chicken nuget

        Results

          Food Code       Name                        Quantity      Price       Canteen Name
        ---------------------------------------------------------------------------------------

No results found for food 'chicken'
```

Figure - 4.15

```
        Search by Food

Enter food name: Chicken

        Results

          Food Code       Name                      Quantity      Price       Canteen Name
        ---------------------------------------------------------------------------------------
           1      |  Chicken Burger              |    20      |  50.00     BUBT Building-3 Canteen
        ---------------------------------------------------------------------------------------
           2      |  Chicken Roll                |    13      |  35.00     BUBT Building-3 Canteen
        ---------------------------------------------------------------------------------------
           3      |  Chicken nugget              |    34      |  40.00     BUBT Building-3 Canteen
        ---------------------------------------------------------------------------------------
           4      |  Chicken Roll                |    22      |  45.00     BUBT BBA Canteen
        ---------------------------------------------------------------------------------------
        Enter item number to add to cart (enter 0 to cancel): 3
Enter quantity: 10
Item added to cart

Enter item number to add to cart (enter 0 to cancel): 4
Enter quantity: 2
Item added to cart

Enter item number to add to cart (enter 0 to cancel):
```

Figure - 4.16

13. After finalizing the order, now I select the credit payment method. This requires me to enter my card number and pin number. After that, my order will be confirmed successfully. Then I have to select option 5 to exit.

```
Do you want to confirm your order?
        1) Yes
        2) No
Enter your choice: 1
        Enter your building number(B1/ B2/ B3/ B4) : B4
        Enter your room number : 603

Payment Method:
        1.Cash
        2.Credit
        2

Enter your card number: 123456

Enter your pin number: ****


The order has been placed successfully.
```

Figure - 4.17

14. Now I will select admin panel and enter admin password to see option like Figure - 4.18. Option 1 will show the menu cards of 2 canteens.

```
Enter admin password: password123


        Admin Panel

Please select an option:

        1) Show Menu
        2) Add Food
        3) Delete Food
        4) Show Instant Order Preview
        5) Show Total Cash Today
        6) Logout

Enter your choice:
```

Figure - 4.18

15. If admin select option 2 then he can add the food, price, and quantity of the food item. Also, he can add the canteen where he wants to add the food. If there is already available that food item, then the admin can add another type of food or he can change the price and quantity.

```
Enter the name of the food item: Water
Enter the price of the food item: 15
Enter the quantity of the food item: 40
Enter the canteen for the food item (1/2): 1
The food item has been added to BUBT Building-3 Canteen.
```

Figure - 4.19

```
Enter the name of the food item: Noodles
Enter the price of the food item: 50
Enter the quantity of the food item: 20
Enter the canteen for the food item (1/2): 1
Noodles is already in BUBT Building-3 Canteen menu.
Do you want to
   1)add another Noodles in BUBT Building-3 Canteen menu
   2)update the price and quantity of Noodles in BUBT Building-3 Canteen menu
   3)Back
:2

        Menu has been updated.
```

Figure - 4.20

16. If the admin selects option 3 then he can delete the food. If the food item doesn't available in the canteen then there will be an error. In Figure - 4.23, we can see which item was added and which item was deleted.

```
Enter the name of the food item to delete: Pizza
Enter the canteen for the food item (1/2): 2
The food item does not exist in the selected canteen.
```

Figure - 4.21

```
Enter the name of the food item to delete: Pizza
Enter the canteen for the food item (1/2): 1
The food item has been deleted.
```

Figure - 4.22

```
            Welcome to BUBT Building-3 Canteen

  Food Code       Name                          Quantity      Price
-------------------------------------------------------------------------
   1        |  Chicken Burger                 |    20      |  50.00
-------------------------------------------------------------------------
   2        |  Burger                         |    16      |  55.00
-------------------------------------------------------------------------
   3        |  Chicken Roll                   |    22      |  35.00
-------------------------------------------------------------------------
   4        |  Sandwich                       |    25      |  45.00
-------------------------------------------------------------------------
   5        |                                 |     0      |  0.00
-------------------------------------------------------------------------
   6        |  Chicken nugget                 |    34      |  40.00
-------------------------------------------------------------------------
   7        |  Ice cream                      |    24      |  50.00
-------------------------------------------------------------------------
   8        |  Noodles                        |    20      |  50.00
-------------------------------------------------------------------------
   9        |  Chips                          |    20      |  20.00
-------------------------------------------------------------------------
  10        |  French Fries                   |    15      |  100.00
-------------------------------------------------------------------------
  11        |  Water                          |    40      |  15.00
-------------------------------------------------------------------------
```

Figure - 4.23

17. If admin select option 4 then he can see the recent orders.

```
      Recent Orders
      --------------

Order - 1

        Food Code       Name                        Quantity      Price
      ------------------------------------------------------------------------
         1        |  Chicken Roll                  |     9      |  315.00
      ------------------------------------------------------------------------
         2        |  Sandwich                      |     5      |  225.00
      ------------------------------------------------------------------------
         3        |  Ice cream                     |    10      |  500.00
      ------------------------------------------------------------------------
         4        |  Khichuri                      |     4      |  220.00
      ------------------------------------------------------------------------
         5        |  Cake                          |    10      |  100.00
      ------------------------------------------------------------------------
         6        |  Club Sandwich                 |     9      |  720.00
      ------------------------------------------------------------------------
        Total                                                 : 2080.00
      ------------------------------------------------------------------------
        Discount: 208.00
      ------------------------------------------------------------------------
        Total bill (after discount): 1872.00
      ------------------------------------------------------------------------
        Building : B2
      ------------------------------------------------------------------------
        Room number : 505
      ------------------------------------------------------------------------
```

Figure - 4.24

```
Order - 2

        Food Code       Name                        Quantity      Price
      ------------------------------------------------------------------------
         1        |  Burger                        |     4      |  200.00
      ------------------------------------------------------------------------
         2        |  Sandwich                      |     5      |  225.00
      ------------------------------------------------------------------------
         3        |  French Fries                  |     3      |  300.00
      ------------------------------------------------------------------------
         4        |  Chicken nugget                |    10      |  400.00
      ------------------------------------------------------------------------
         5        |  Chicken Roll                  |     2      |  90.00
      ------------------------------------------------------------------------
        Total                                                 : 1215.00
      ------------------------------------------------------------------------
        Discount: 121.50
      ------------------------------------------------------------------------
        Total bill (after discount): 1093.50
      ------------------------------------------------------------------------
        Building : B4
      ------------------------------------------------------------------------
        Room number : 603
      ------------------------------------------------------------------------
```

Figure - 4.25

18. If admin select option 5 then he can see today's total cash like figure- 4.26.

```
Total cash earned today: 3295.00
Total discount applied today: 329.50
Net cash earned today: 2965.50
```

Figure - 4.26

19. After seeing the today's total cash admin will select the option 5 for logout. Then select option 3 to exit. After exit there will be a message like Figure - 4.28.

```
Logged out successfully...

Please select an option:

        1) Customer
        2) Admin Panel
        3) Exit

Enter your choice:
```

Figure - 4.27

```
Goodbye!


Process returned 0 (0x0)   execution time : 24811.514 s
Press any key to continue.
```

Figure - 4.28

Chapter 5

Code Analysis

5.1 - Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <windows.h>
#define MAX_CART_SIZE 100
#define MAX_USERS 100
#define MAX_FOOD_ITEMS 20
#define MAX_ORDER_ITEMS 50
#define ADMIN_PASSWORD "password123"

void cls(){
	system("cls");
}

// Define user structure
typedef struct {
	char name[50];
	int id;
	char email[50];
	char password[50];
	char mobile[20];
	float total_bill;
	float discount;
} User;

// Define food structure
typedef struct {
	char cname[30];
	char name[50];
	int quantity;
	float price;
	float discount;
	int count;
	char build[3];
	int room;
} Food;
```

```
Food canteen1[MAX_FOOD_ITEMS] = {
                                {"BUBT Building-3 Canteen","Chicken Burger",20,50},
                                {"BUBT Building-3 Canteen","Burger",16, 55},
                                {"BUBT Building-3 Canteen","Chicken Roll",22, 35},
                                {"BUBT Building-3 Canteen","Sandwich",25, 45},
                                {"BUBT Building-3 Canteen","Pizza",19, 70},
                                {"BUBT Building-3 Canteen","Chicken nugget",34, 40},
                                {"BUBT Building-3 Canteen","Ice cream",24, 50},
                                {"BUBT Building-3 Canteen","Noodles",16, 55},
                                {"BUBT Building-3 Canteen","Chips",20, 20},
                                {"BUBT Building-3 Canteen","French Fries",15, 100}};

Food canteen2[MAX_FOOD_ITEMS] = {
                                {"BUBT BBA Canteen","Burger",20,50},
                                {"BUBT BBA Canteen","Khichuri",16, 55},
                                {"BUBT BBA Canteen","Chicken Roll",22, 45},
                                {"BUBT BBA Canteen","Sandwich",25, 45},
                                {"BUBT BBA Canteen","Pizza(Family size)",18, 70},
                                {"BUBT BBA Canteen","Ice cream",26, 45},
                                {"BUBT BBA Canteen","Beef Burger",19, 50},
                                {"BUBT BBA Canteen","French Fries",19, 100},
                                {"BUBT BBA Canteen","Club Sandwich",20, 80},
                                {"BUBT BBA Canteen","Cake",35, 10}};

// Define global variables
User users[MAX_USERS];
int user_count = 0;
Food menu[MAX_FOOD_ITEMS];
int menu_count = 0;
Food order1[MAX_ORDER_ITEMS];
int order_count = 0;
Food cart[MAX_CART_SIZE];
int cart_count = 0;
float total_bill = 0;
float discount =0;
int a=0;

// Function prototypes
void print_customer_menu();
void handle_customer_input(int input);
void signup();
int login();
```

```
void Canteen1_initialize();
void Canteen2_initialize();
void search_by_canteen();
void search_by_food();
void add_to_cart(Food food, int quantity);
void show_cart();
void payment_method(int choice);
void finalize_order();
void print_admin_menu();
void handle_admin_input(int input);
void show_menu();
void add_food();
void delete_food();
void show_instant_order_preview();
void show_total_cash_today();

int main() {
	system("COLOR B5");
	FILE *fp;
	fp = fopen("users.dat", "rb");
	if (fp != NULL) {
		User user;
		while (fread(&user, sizeof(User), 1, fp)) {
			users[user_count] = user;
			user_count++;
		}
		fclose(fp);
	}
	int choice;
	printf("\n\t********Welcome to BUBT Canteen Food Ordering
System(CFOS)*******\n\n");
	do {
		Sleep(200);
		printf("Please select an option:\n\n");
		Sleep(200);
		printf("\t1) Customer\n");
		Sleep(200);
		printf("\t2) Admin Panel\n");
		Sleep(200);
		printf("\t3) Exit\n\n");
		Sleep(200);
		printf("Enter your choice: ");
```

```
        while (scanf("%d", &choice) != 1 || (choice != 1 && choice != 2 && choice != 3)) {
            printf("Invalid input. Please enter 1 or 2 or 3: ");
            scanf("%*[^\n]");    // clear input buffer
        }
        cls();
        switch (choice) {
            case 1:
                print_customer_menu();
                break;
            case 2:
                printf("\nEnter admin password: ");
                char admin_password[50];
                scanf("%s",admin_password);
                /*int i;
                char c=' ';
                while(i<20)
                {
                    admin_password[i]=getch();
                    c=admin_password[i];
                    if(c=='\r') break;
                    else printf("*");
                    i++;
                }
                admin_password[i]='\0';*/
                if (strcmp(admin_password, ADMIN_PASSWORD) == 0) {
                    print_admin_menu();
                } else {
                    printf("\nIncorrect password\n\n");
                }
                break;
            case 3:
                printf("\nGoodbye!\n\n");
                exit(0);
            default:
                printf("\nInvalid choice. Please enter a valid option.\n\n");
        }
    } while (1);

    return 0;
}
void print_customer_menu() {
    int input;
```

```
do {
        printf("\n\t****Customer Menu****\n\n");
        printf("Please select an option:\n\n");
        Sleep(200);
        printf("\t1) Signup\n");
        Sleep(200);
        printf("\t2) Login\n");
        Sleep(200);
        printf("\t3) Back\n\n");
        Sleep(200);
        printf("Enter your choice: ");
        while (scanf("%d", &input) != 1 || (input != 1 && input != 2 && input != 3)) {
            printf("Invalid input. Please enter 1 or 2 or 3: ");
            scanf("%*[^\n]");    // clear input buffer
        }
        cls();
        handle_customer_input(input);

    } while (input != 3);
}
void handle_customer_input(int input) {
    switch (input) {
        case 1:
            signup();
            break;
        case 2:
            if (login()!=-1) {
                int choice;
                cls();
                do {
                    printf("\n****Search Menu****\n\n");
                    printf("Please select an option:\n\n");
                    Sleep(200);
                    printf("1) Search by Canteen\n");
                    Sleep(200);
                    printf("2) Search by Food\n");
                    Sleep(200);
                    printf("3) Show Cart\n");
                    Sleep(200);
                    printf("4) Finalize Order\n");
                    Sleep(200);
                    printf("5) Logout\n\n");
                    Sleep(200);
```

```
                        printf("Enter your choice: ");

                                while (scanf("%d", &choice) != 1 || (choice != 1 && choice !=
2 && choice != 3 && choice != 4 && choice != 5 )) {
                            printf("Invalid input. Please enter from 1-5: ");
                            scanf("%*[^\n]");    // clear input buffer
                        }

                        cls();
                        switch (choice) {
                            case 1:
                                search_by_canteen();
                                break;
                            case 2:
                                search_by_food();
                                break;
                            case 3:
                                show_cart();
                                break;
                            case 4:
                                finalize_order();
                                break;
                            case 5:
                                printf("\nLogged out successfully\n\n");
                                return;
                            default:
                                printf("\nInvalid choice. Please enter a valid option.\n\n");
                        }
                    } while (1);
                } else {
                    printf("\nIncorrect email or password\n\n");
                }
                break;
            case 3:
                printf("\nGoing back to main menu...\n\n");
                return;
            default:
                printf("\nInvalid choice. Please enter a valid option.\n\n");
        }
}

void signup(){
    cls();
```

```
    printf("\n\t****Signup****\n\n");
    User user;
    while (1) {
        printf("Enter your name: ");
        fflush(stdin);
        fgets(user.name, 50, stdin);
        user.name[strcspn(user.name, "\n")] = 0;


            int valid = 1;
        for (int i = 0; i < strlen(user.name); i++) {
            if (!isalpha(user.name[i]) && !isspace(user.name[i])) {
                valid = 0;
                break;
            }
        }
        if (valid) {
            break;
        }
        printf("\tInvalid name! Please enter a name with alphabets only.\n\t");
    }
    // Get user's email, checking for @ and .
    while (1) {
        printf("Enter your email: ");
        fflush(stdin);
        fgets(user.email, 50, stdin);
        user.email[strcspn(user.email, "\n")] = 0;

        if (strlen(user.email) < 5 || strchr(user.email, '@') == NULL || strchr(user.email, '.') ==
NULL) {
            printf("Invalid email!(Email must contain @, a dot and length should be greater
than 5.)\n");
        } else {
            break;
        }
    }
    // Get user's mobile, checking for digits only
    while (1) {
        printf("Enter your mobile number: ");
        fflush(stdin);
        fgets(user.mobile, 20, stdin);
        user.mobile[strcspn(user.mobile, "\n")] = 0;

        int valid = 1;
```

```
        for (int i = 0; i < strlen(user.mobile); i++) {
            if (!isdigit(user.mobile[i])) {
                valid = 0;
                break;
            }
        }
        if (valid && strlen(user.mobile) == 11) {
            break;
        } else {
            printf("Invalid mobile number! Please enter a valid 11-digit number.\n");
        }
    }
    // Get user's password, checking for length and complexity
    while (1) {
        printf("Enter a password: ");
        fflush(stdin);
        fgets(user.password, 50, stdin);
        user.password[strcspn(user.password, "\n")] = 0;

        int valid = 1;
        if (strlen(user.password) < 5 || strlen(user.password) > 12) {
            valid = 0;
            printf("Password length must be between 5 to 12 characters.\n");
        } else {
            for (int i = 0; i < strlen(user.password); i++) {
                    if (!isupper(user.password[i]) && !islower(user.password[i]) &&
!isdigit(user.password[i]) && isalnum(user.password[i])) {
                        valid = 0;
                        printf("Password must contain at least one uppercase letter, one
lowercase letter, one digit, and one special character.\n");
                }
            }
        }
        if (valid) {
            break;
        }
        printf("Invalid password! Please enter a valid password.\n");
    }
    user.id = user_count + 1;
    user.total_bill = 0.0;
    user.discount = 0.0;

    // Save user information to file
```

```
	FILE *fp;
	fp = fopen("users.dat", "ab");
	if (fp == NULL) {
		printf("Error opening file!\n");
		return;
	}
	fwrite(&user, sizeof(User), 1, fp);
	fclose(fp);

	users[user_count] = user;
	user_count++;

	printf("\n\nSignup successful! Please login to continue.\n");
	printf("\n\tPress any key to continue...");
	getch();
}

int login() {
	char email[50], password[50];
	printf("\n\t****Login****\n\n");
	printf("Enter your email: ");
	scanf("%s", email);
	printf("Enter your password: ");
	scanf("%s", password);
	for (int i = 0; i < user_count; i++) {
		if (strcmp(users[i].email, email) == 0 && strcmp(users[i].password, password) == 0) {
			printf("\nLogin successful. Welcome %s!\n\n", users[i].name);
			Sleep(900);
			return i;
		}
	}
	return -1;
}

void Canteen1_initialize(){
			printf("\n\t\tWelcome to BUBT Building-3 Canteen\t\t");
			printf("\n\n\t	Food Code		Name					Quantity
Price		");Sleep(200);
			printf("\n\t");
			printf("----------------------------------------------------------------------");Sleep(200);
			for (int i = 0; i < MAX_FOOD_ITEMS; i++) {
				printf("\n\t");
```

```
                    printf("    %2d              |    %-25s          |        %2d          |    %0.2f      ",i+1,
canteen1[i].name, canteen1[i].quantity, canteen1[i].price);
                    printf("\n\t");
                    printf("----------------------------------------------------------------------");
                }
                     printf("\n\n");
}
void Canteen2_initialize(){
                printf("\n\t\tWelcome to BUBT BBA Canteen\t\t");
                printf("\n\n\t    Food Code              Name                                    Quantity
Price            ");Sleep(200);
                printf("\n\t");
                printf("----------------------------------------------------------------------");Sleep(200);
                for (int i = 0; i < MAX_FOOD_ITEMS; i++) {
                    printf("\n\t");
                    printf("    %2d            |    %-25s          |        %2d          |    %0.2f      ",i+1,
canteen2[i].name, canteen2[i].quantity, canteen2[i].price);
                    printf("\n\t");
                    printf("----------------------------------------------------------------------");
                }
                printf("\n\n");
}
void search_by_canteen() {
    printf("\n\tSearch by Canteen\n\n");
    int input;
    printf("Please select an option:\n");
    Sleep(200);
    printf("\t1) Building-3 Canteen\n");
    Sleep(200);
    printf("\t2) BBA Canteen\n");
    Sleep(200);
    printf("\t3) Back\n");
    Sleep(200);
    printf("Enter canteen number: ");
    /*scanf("%d",&input);*/
    while (scanf("%d", &input) != 1 || (input != 1 && input != 2 && input != 3)) {
        printf("Invalid input. Please enter 1 or 2 or 3: ");
        scanf("%*[^\n]");    // clear input buffer
    }

    switch(input){
        case 1:
            Canteen1_initialize();
```

```
                break;
            case 2:
                Canteen2_initialize();
                break;
            case 3:
                return;
            default:
                printf("\nPlease Enter the valid choice\n\n");
                return;
        }
          int item_choice;
        int d=1;
        while(d!=0){
            printf("Enter item number to add to cart (enter 0 to cancel): ");
            scanf("%d", &item_choice);
            if (item_choice < 0 || item_choice > MAX_FOOD_ITEMS) {
                printf("\nInvalid choice. Please enter a valid item number.\n");
            } else if (item_choice == 0) {
                cls();
                return;
            } else {
                int qunti;
                printf("Enter quantity: ");
                scanf("%d", &qunti);
                if (qunti <= 0) {
                    printf("\nQuantity should be greater than zero.\n");
                } else if (order_count + qunti > MAX_CART_SIZE) {
                    printf("\nCannot add more items. Cart size limit exceeded.\n");
                } else {
                    if(input==1){
                        add_to_cart(canteen1[item_choice-1], qunti);
                    }
                    else if(input==2){
                        add_to_cart(canteen2[item_choice-1], qunti);
                    }
                }
            }
        }
}
void search_by_food() {
        printf("\n\tSearch by Food\n\n");
        char food[50];
        printf("Enter food name: ");
```

```
    scanf("%s",food);
    printf("\n\tResults");
    int count = 0;
    printf("\n\n\t    Food Code            Name                                Quantity          Price
Canteen Name");Sleep(200);
    printf("\n\t");
    printf("-----------------------------------------------------------------------------------------
");Sleep(200);
    printf("\n\t");
    for (int i = 0; i < MAX_FOOD_ITEMS; i++) {
        if (strstr(canteen1[i].name, food) != NULL) {
            printf("    %2d          |   %-25s     |       %2d          |   %0.2f         %s", count+1,
canteen1[i].name, canteen1[i].quantity, canteen1[i].price,canteen1->cname);
            printf("\n\t");
            printf("-----------------------------------------------------------------------------------------");
            printf("\n\t");
            menu[count]=canteen1[i];
            count++;
        }
    }
    for (int i = 0; i < MAX_FOOD_ITEMS; i++) {
        if (strstr(canteen2[i].name, food) != NULL) {
            printf("    %2d          |   %-25s     |       %2d          |   %0.2f         %s", count+1,
canteen2[i].name, canteen2[i].quantity, canteen2[i].price,canteen2->cname);
            printf("\n\t");
            printf("-----------------------------------------------------------------------------------------");
            printf("\n\t");
            menu[count]=canteen2[i];
            count++;
        }
    }
    if (count== 0) {
        printf("\nNo results found for food '%s'\n", food);
    } else {
        int item_choice;
        int d=1;
        while(d!=0){
            printf("Enter item number to add to cart (enter 0 to cancel): ");
            if (scanf("%d", &item_choice) != 1) {
                printf("\nInvalid input. Please enter a numeric item number.\n");
                cls(); // Assuming cls() is a function to clear screen
                return;
            }
```

```
                if (item_choice < 0 || item_choice > count) {
                        printf("\nInvalid choice. Please enter a valid item number.\n");
                } else if (item_choice == 0) {
                        printf("\nCancelled\n");
                        cls();
                        return;
                } else {
                        int qunty;
                        printf("Enter quantity: ");
                        scanf("%d", &qunty);
                        if (qunty <= 0) {
                                printf("\nQuantity should be greater than zero.\n");
                        } else if (order_count + qunty > MAX_CART_SIZE) {
                                printf("\nCannot add more items. Cart size limit exceeded.\n");
                        } else if (qunty > menu[item_choice-1].quantity) {
                                printf("\nQuantity exceeds available quantity.\n");
                        } else {
                                add_to_cart(menu[item_choice-1], qunty);
                        }
                }
        }
    }
}
// Function to add a food item to the cart
void add_to_cart(Food food, int quantity) {
    // Check if the cart is already full
    if (cart_count == MAX_CART_SIZE) {
        printf("Sorry, the cart is full.\n");
        return;
    }
    // Check if there is enough quantity of the selected food item
    if (quantity > food.quantity) {
        printf("Sorry, there is not enough quantity of the selected food item.\n\n");
        return;
    }
    // Check if the quantity entered is valid
    if (quantity <= 0) {
        printf("Please enter a valid quantity.\n\n");
        return;
    }

    // Add the food item to the cart
    cart[cart_count] = food;
```

```
	cart[cart_count].quantity = quantity;
	total_bill += food.price * quantity;
	food.quantity -= quantity;
	cart_count++;
	printf("Item added to cart\n\n");
}

void show_cart() {
	float total = 0;
	printf("\n\tCart Contents:\n");
	printf("\n\n\t	Name				Quantity		Price
Total");Sleep(200);
	printf("\n\t");
	printf("-----------------------------------------------------------------------");Sleep(200);
	printf("\n\t");
	for (int i = 0; i < cart_count; i++) {
		printf("	%-25s	|	%2d		|	%0.2f	|	%0.2f	",	cart[i].name,
cart[i].quantity, cart[i].price, cart[i].price * cart[i].quantity);
		printf("\n\t");
		printf("-----------------------------------------------------------------------");
		printf("\n\t");
		total += cart[i].price * cart[i].quantity;
	}
	printf("	Total								: %0.2f\n",
total);
	printf("\t-----------------------------------------------------------------------\n");Sleep(200);
	// Calculate discounted total bill, if applicable
	if(total_bill >= 300 && total_bill < 500) {
		discount = (total_bill * 0.05);
		printf("\t	Discount: %0.2f\n",discount);
		printf("\t-----------------------------------------------------------------------\n");
		printf("\t	Total bill (after 5%% discount): %0.2f\n", total_bill-discount);
		printf("\t-----------------------------------------------------------------------\n\n");
	}
	else if(total_bill >= 500) {
		discount = (total_bill * 0.1);
		printf("\t	Discount: %0.2f\n",discount);
		printf("\t-----------------------------------------------------------------------\n");
		printf("\t	Total bill (after 10%% discount): %0.2f\n", total_bill-discount);
		printf("\t-----------------------------------------------------------------------\n\n");
	}
	else{
		discount =0.0;
```

```
            printf("\n");
        }
}

void payment_method(int choice){
        int num;
        char pin[20];
        switch(choice){
        case 1:
            printf("\n\t=========>THANK YOU<=========");
            break;
        case 2:
            printf("\nEnter your card number: ");
            scanf("%d",&num);
            printf("\nEnter your pin number: ");
            int i;
                        char c=' ';
                        while(i<20)
                        {
                            pin[i]=getch();
                            c=pin[i];
                            if(c=='\r') break;
                            else printf("*");
                            i++;
                        }
                        pin[i]='\0';
        }
}

// Function to finalize the order and update the quantity of food items
void finalize_order() {
        char bui[3];
        int b;
        int c;
        // Check if the cart is empty
        if (cart_count == 0) {
            printf("The cart is empty.\n");
            return;
        }

        show_cart();

        int confirm;
```

```
    printf("\nDo you want to confirm your order?\n");
    printf("\t1) Yes\n");
    printf("\t2) No\n");
    printf("Enter your choice: ");
    scanf("%d", &confirm);
    if(confirm==1){
        printf("\tEnter your building number(B1/ B2/ B3/ B4) : ");
        /*scanf("%s", bui);*/
        while (scanf("%s", bui) != 1 || (strcmp(bui , "B1")!=0 && strcmp(bui , "B2")!=0) &&
strcmp(bui , "B3")!=0 && strcmp(bui , "B4")!=0) {
            printf("Invalid input. Please enter B1/ B2/ B3/ B4: ");
            scanf("%*[^\n]");    // clear input buffer
        }
        printf("\tEnter your room number : ");
        while (scanf("%d", &b) != 1 ) {
            printf("Invalid input. Please enter valid room number : ");
            scanf("%*[^\n]");    // clear input buffer
        }
        printf("\nPayment Method:\n");
        printf("\t1.Cash\n\t2.Credit\n\t");
        while (scanf("%d", &c) != 1 || (c != 1 && c != 2)) {
            printf("Invalid input. Please enter 1 or 2: ");
            scanf("%*[^\n]");    // clear input buffer
        }
        payment_method(c);

        // Update the quantity of food items and order history
        for (int i = 0; i < cart_count; i++) {
            for (int j = 0; j < MAX_FOOD_ITEMS; j++) {
                if (strcmp(cart[i].name, canteen1[j].name) == 0) {
                    if (cart[i].quantity > canteen1[j].quantity) {
                        printf("Sorry, there is not enough quantity of %s in %s.\n",
cart[i].name,canteen1->cname);
                        return;
                    }
                    canteen1[j].quantity -= cart[i].quantity;
                }
                else if (strcmp(cart[i].name, canteen2[j].name) == 0) {
                    if (cart[i].quantity > canteen2[j].quantity) {
                        printf("Sorry, there is not enough quantity of %s in %s.\n",
cart[i].name,canteen2->cname);
                        return;
                    }
```

```
                        canteen2[j].quantity -= cart[i].quantity;
                    }
                }
                order1[a].count=cart_count;
                order1[order_count] = cart[i];
                order1[a].discount = discount;
                strcpy(order1[a].build , bui);
                order1[a].room = b;
                order_count++;
            }
            // Reset the cart and total bill
            memset(cart, 0, sizeof(cart));
            cart_count = 0;
            total_bill = 0;
            a++;
            printf("\n\n\nThe order has been placed successfully.\n");
        }else{
            cls();
            return;
        }
}

void print_admin_menu() {
        int input;
        do {
            printf("\n\n\tAdmin Panel\n\n");
            printf("Please select an option:\n\n");
            Sleep(200);
            printf("\t1) Show Menu\n");
            Sleep(200);
            printf("\t2) Add Food\n");
            Sleep(200);
            printf("\t3) Delete Food\n");
            Sleep(200);
            printf("\t4) Show Instant Order Preview\n");
            Sleep(200);
            printf("\t5) Show Total Cash Today\n");
            Sleep(200);
            printf("\t6) Logout\n\n");
            Sleep(200);
            printf("Enter your choice: ");
            while (scanf("%d", &input) != 1 || (input != 1 && input != 2 && input != 3 && input
!= 4 && input != 5 && input != 6)) {
```

```
                printf("Invalid input. Please enter 1-6: ");
                scanf("%*[^\n]");    // clear input buffer
            }
            cls();
            handle_admin_input(input);

        } while (input != 6);
}

void handle_admin_input(int input) {
        switch (input) {
            case 1:
                show_menu();
                break;
            case 2:
                add_food();
                break;
            case 3:
                delete_food();
                break;
            case 4:
                show_instant_order_preview();
                break;
            case 5:
                show_total_cash_today();
                break;
            case 6:
                printf("\nLogged out successfully...\n\n");
                return;
            default:
                printf("\nInvalid choice. Please enter a valid option.\n\n");
        }
}
void show_menu() {
        printf("\n\tMenu\n\n");
        Canteen1_initialize();
        Canteen2_initialize();
}

void add_food() {
        char name[50];
        float price;
        int quantity, canteen_choice;
```

```
    printf("Enter the name of the food item: ");
    scanf("%s", name);

    printf("Enter the price of the food item: ");
    while (scanf("%f", &price) != 1 || price < 0) {
        printf("Invalid input. Please enter a non-negative price: ");
        scanf("%*[^\n]");    // clear input buffer
    }

    printf("Enter the quantity of the food item: ");
    while (scanf("%d", &quantity) != 1 || quantity < 0) {
        printf("Invalid input. Please enter a non-negative quantity: ");
        scanf("%*[^\n]");    // clear input buffer
    }

    printf("Enter the canteen for the food item (1/2): ");
    while (scanf("%d", &canteen_choice) != 1 || (canteen_choice != 1 && canteen_choice !=
2)) {
        printf("Invalid input. Please enter 1 or 2: ");
        scanf("%*[^\n]");    // clear input buffer
    }

    Food* canteen;
    if (canteen_choice == 1) {
        canteen = canteen1;
    } else {
        canteen = canteen2;
    }
    int choice;
    for (int i = 0; i < MAX_FOOD_ITEMS; i++) {
        if (strcmp(name, canteen[i].name) == 0) {
            printf("%s is already in %s menu.\n", name,canteen->cname);
            printf("Do you want to \n      1)add another %s in %s menu \n      2)update the
price and quantity of %s in %s menu\n      3)Back\n:",name,canteen->cname,name,canteen-
>cname);
            while (scanf("%d", &choice) != 1 || (choice != 1 && choice != 2 && choice !=
3)) {
                printf("Invalid input. Please enter 1 or 2 or 3: ");
                scanf("%*[^\n]");    // clear input buffer
            }
            if(choice==1){
                for (int i = 0; i < MAX_FOOD_ITEMS; i++) {
```

```
                        if (strlen(canteen[i].name) == 0) {
                            strcpy(canteen[i].name, name);
                            canteen[i].price = price;
                            canteen[i].quantity = quantity;
                            printf("\nThe food item has been added to %s.\n", canteen-
>cname);
                            return;
                        }
                    }
                    printf("\nSorry, %s menu is already full.\n", canteen->cname);
                    return;
                }
                else if(choice==2){
                    canteen[i].price=price;
                    canteen[i].quantity=quantity;
                    printf("\n\tMenu has been updated.\n");
                    return;
                }
                else{
                    return;
                }
            }
        }
        for (int i = 0; i < MAX_FOOD_ITEMS; i++) {
            if (strlen(canteen[i].name) == 0) {
                strcpy(canteen[i].name, name);
                canteen[i].price = price;
                canteen[i].quantity = quantity;
                printf("The food item has been added to %s.\n", canteen->cname);
                return;
            }
        }
        printf("Sorry, %s menu is already full.\n", canteen->cname);
}
void delete_food() {
        char name[50];
        int canteen_choice;

        printf("Enter the name of the food item to delete: ");
        scanf("%s", name);

        // Check if the food item exists in any canteen
        int item_found = 0;
```

```
    for (int i = 0; i < MAX_FOOD_ITEMS; i++) {
        if (strcmp(canteen1[i].name, name) == 0 || strcmp(canteen2[i].name, name) == 0) {
            item_found = 1;
            break;
        }
    }
    if (!item_found) {
        printf("The food item does not exist in any canteen.\n");
        return;
    }
    printf("Enter the canteen for the food item (1/2): ");

    while (scanf("%d", &canteen_choice) != 1 || (canteen_choice != 1 && canteen_choice !=
2)) {
        printf("Invalid input. Please enter 1 or 2: ");
        scanf("%*[^\n]");    // clear input buffer
    }

    // Delete the food item from the selected canteen
    Food* canteen;
    if (canteen_choice == 1) {
        canteen = canteen1;
    } else {
        canteen = canteen2;
    }
    int item_deleted = 0;
    for (int i = 0; i < MAX_FOOD_ITEMS; i++) {
        if (strcmp(canteen[i].name, name) == 0) {
            memset(&canteen[i], 0, sizeof(canteen[i]));
            printf("The food item has been deleted.\n");
            item_deleted = 1;
            break;
        }
    }
    if (!item_deleted) {
        printf("The food item does not exist in the selected canteen.\n");
    }
}

// Function to show instant order preview
void show_instant_order_preview() {
    int i,j,k,n=0;
    float total=0;
```

```
    // Print header
    printf("\n\tRecent Orders\n");
    printf("\t--------------");
    for(i = 0 ; i < a ; i++){
        printf("\n\n    Order - %d\n",i+1);
        printf("\n\t    Food Code            Name                                    Quantity
Price           ");
        printf("\n\t");
        printf("----------------------------------------------------------------------");
        // Print orders
        for (j = n, k = 0; k < order1[i].count; j++, k++) {
            printf("\n\t    %2d              |    %-25s          |       %2d           |    %0.2f      ", k+1
, order1[j].name, order1[j].quantity, order1[j].price*order1[j].quantity);
            printf("\n\t");
            printf("----------------------------------------------------------------------");
            total += (order1[j].price * order1[j].quantity);
        }
        n += k;
        printf("\n\t");
        printf("    Total                                                                    : %0.2f\n",
total);
        printf("\t----------------------------------------------------------------------\n");
        // Calculate discounted total bill, if applicable
        if(order1[i].discount !=0) {
            printf("\t    Discount: %0.2f\n",order1[i].discount);
            printf("\t----------------------------------------------------------------------\n");
            printf("\t    Total bill (after discount): %0.2f\n", total-order1[i].discount);
            printf("\t----------------------------------------------------------------------\n");
        }
        printf("\t    Building : %s\n",order1[i].build);
        printf("\t----------------------------------------------------------------------\n");
        printf("\t    Room number : %d\n",order1[i].room);
        printf("\t----------------------------------------------------------------------\n\n");
        total=0;
    }
}

// Function to show total cash earned today
void show_total_cash_today() {
    float total_cash = 0;
    float total_discount = 0;
    float net_cash = 0;
```

```
    // Calculate total cash earned from orders
    for (int i = 0; i < order_count; i++) {
        total_cash += (order1[i].price * order1[i].quantity);
    }

    // Calculate the total discount
    for (int i = 0; i < a; i++) {
        total_discount += order1[i].discount;
    }
    net_cash=total_cash-total_discount;
    printf("\nTotal cash earned today: %.2f\n", total_cash);
    printf("Total discount applied today: %.2f\n", total_discount);
    printf("Net cash earned today: %.2f\n", net_cash);
}
```

Chapter 6

Future plan

6.1 Future Plan

There are several improvements that can be made to the BUBT Canteen Food Ordering System. Firstly, we could add more menu items and provide nutritional information for each item. This would help students make informed decisions about their meals. Secondly, we could integrate the system with a mobile app, which would make it more convenient for students to place orders from their phones. Finally, we could consider adding a loyalty program or other incentives to encourage students to use the system regularly. These improvements would make the system even more user-friendly and help to increase adoption among students.

Chapter 7

Conclusion

7.1 Conclusion

The main aim of this project was to develop our university canteen food system. To save time for students who are waiting in a long line to order their food and another line to receive their delivery, In this system, we have created a programming project that is more time-saving and informative than the normal canteens. This system will help all the University Canteen customers order their food online. Many customers can easily place orders using their mobile phone, computer, and the internet. So we believe this system will help us. Then any university canteen can hire delivery boys, and they will create a new job for them.